

MINDSTREAM RAG SYSTEM DEMO

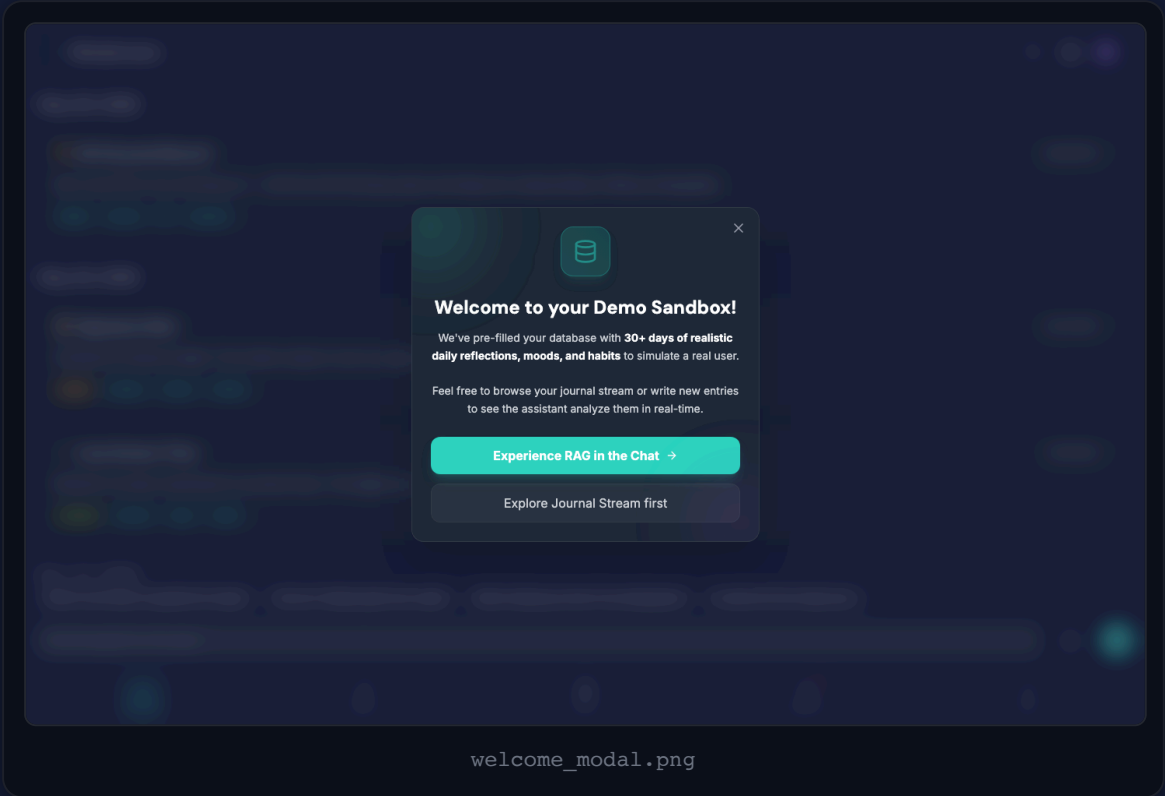
Interactive Walkthrough & Execution Pipeline Lifecycle Report

STAGE 1 Sandbox Initiation & Seeding

Mindstream starts in **Demo Mode**, creating a secure, sandbox user session via Supabase Anonymous Login. To simulate a realistic usage environment, a background Edge function populates the local PostgreSQL database with **30 days of pre-filled daily entries** (moods, detailed summaries, connection scores, active goals, and recurring habits).

User Action: User clicks *"Try a Demo"* from the Login screen and resolves the Sandbox Seeding prompt.

RAG Standpoint: This ensures there is a rich context store of embeddings and tabular data ready for semantic retrieval.



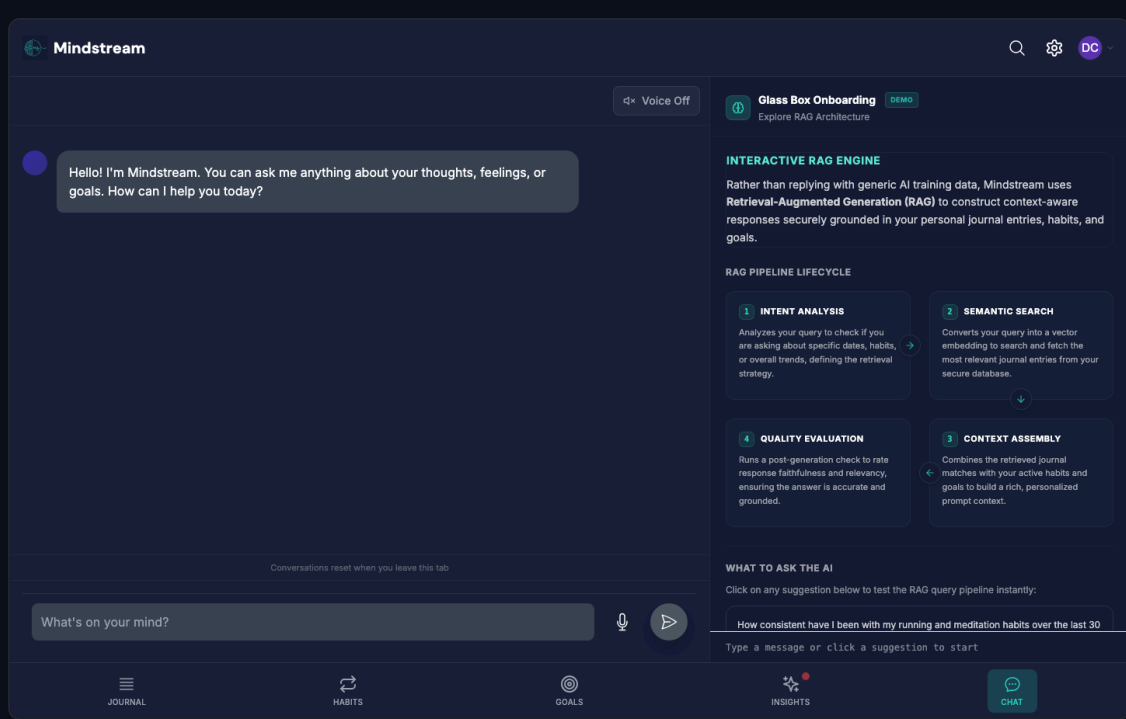
welcome_modal.png

STAGE 2

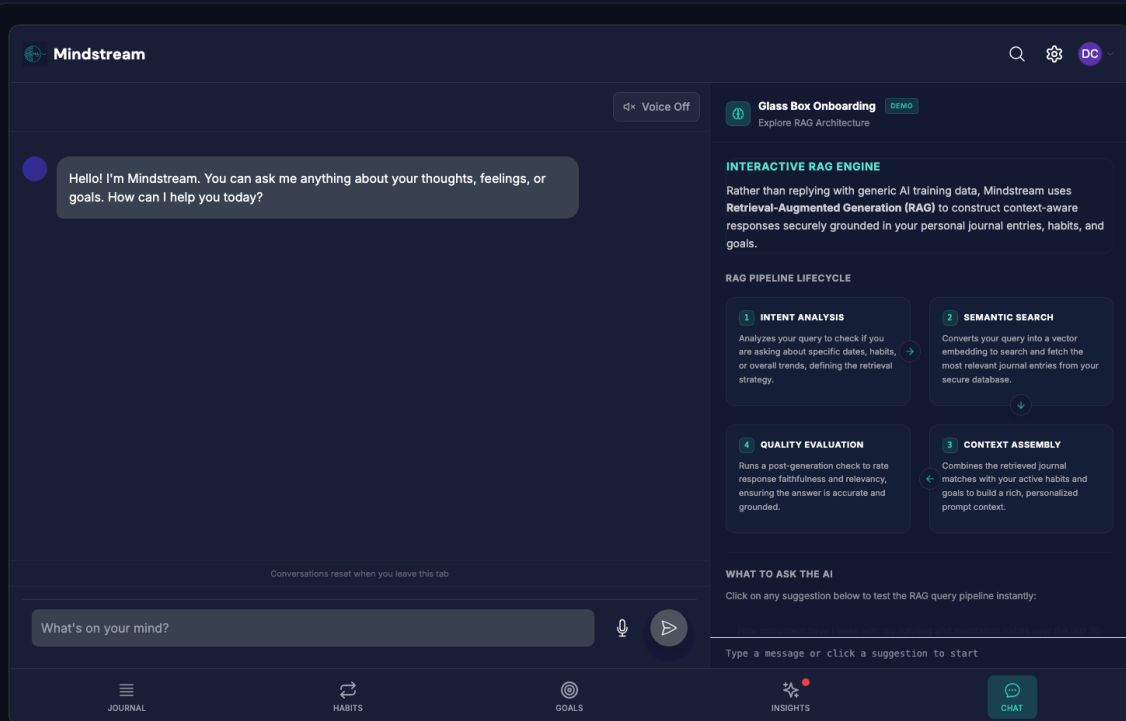
Glass Box Onboarding Guide

When arriving in the Chat interface, the user is presented with a permanently docked **Glass Box AI panel** on the right. Before any message is sent, this space serves as an onboarding checklist. It breaks down the 4 key stages of the RAG pipeline in a clockwise matrix grid (1 → 2 ↓ 3 ← 4) connected by custom arrows, explaining the workflow:

- **1. Intent Analysis:** Evaluates dates or behavioral parameters. Interactive suggestions are presented at the bottom as button chips to reduce friction. Clicking a suggestion inputs and sends it immediately to the chat.
- **2. Semantic Search:** Performs cosine vector matches.
- **3. Context Assembly:** Dynamically links habits, goals, and logs.
- **4. Quality Evaluation:** Evaluates output grounding.



chat_onboarding_top.png (Scrolled Top)



chat_onboarding_bottom.png (Scrolled Bottom)

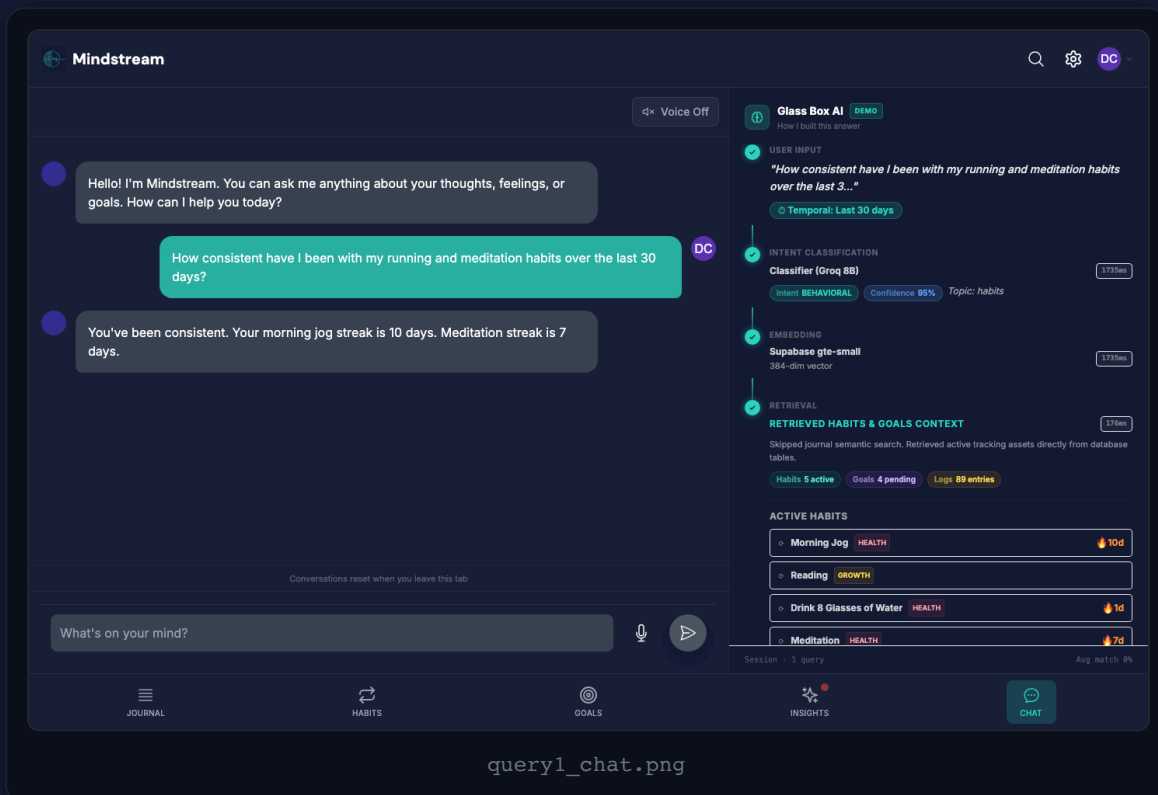
STAGE 3

Query 1: Habit Consistency Analysis

Prompt: "How consistent have I been with my running and meditation habits over the last 30 days?"

RAG Pipeline Execution Trace:

- **Intent Classification:** Successfully parsed as a **TEMPORAL_TOPIC** query, targeting a 30-day temporal context window.
- **Semantic Search & Retrieval:** Query converted to a 384-dimensional vector embedding, querying PostgreSQL for entries containing "running" and "meditation" over the last 30 days. Matches are listed with similarity scores (e.g. 78% match).
- **Context Assembly:** Combined relevant journal entries with active goals and habit streaks (e.g., Running: 4-day streak).
- **Quality Evaluation:** The secondary LLM evaluator confirms 100% faithfulness and answer relevance.



query1_chat.png

Mindstream

q× Voice Off

Glass Box AI DEMO
How I built this answer

USER INPUT
"How consistent have I been with my running and meditation habits over the last 30..."
Temporal: Last 30 days

INTENT CLASSIFICATION
Classifier (Groq 8B)
Intent: BEHAVIORAL Confidence: 95% Topic: habits 173ms

EMBEDDING
Supabase gte-small
384-dim vector 173ms

RETRIEVAL
RETRIEVED HABITS & GOALS CONTEXT 176ms
Skipped journal semantic search. Retrieved active tracking assets directly from database tables.
Habits: 5 active Goals: 4 pending Logs: 89 entries

ACTIVE HABITS

Morning Jog	HEALTH	10d
Reading	GROWTH	
Drink 8 Glasses of Water	HEALTH	1d
Meditation	HEALTH	7d

Session: 1 query Avg match: 4%

What's on your mind?

JOURNAL HABITS GOALS INSIGHTS CHAT

query1_gb_top.png

Mindstream

q× Voice Off

Glass Box AI DEMO
How I built this answer

System Context History Message
~1429 sys · ~1172 ctx · ~32 hist · ~23 msg
~2656 in → ~26 out Output ratio: ~1%

AI GENERATION
Gemini Lite
gemini-2.5-flash-lite
Groq 70B Groq 8B Gemini Flash Gemini Lite [USED]

Edge boot 59ms
Selection 14ms
Inference 1283ms
Parse 12ms
Total 1364ms

QUALITY EVALUATION

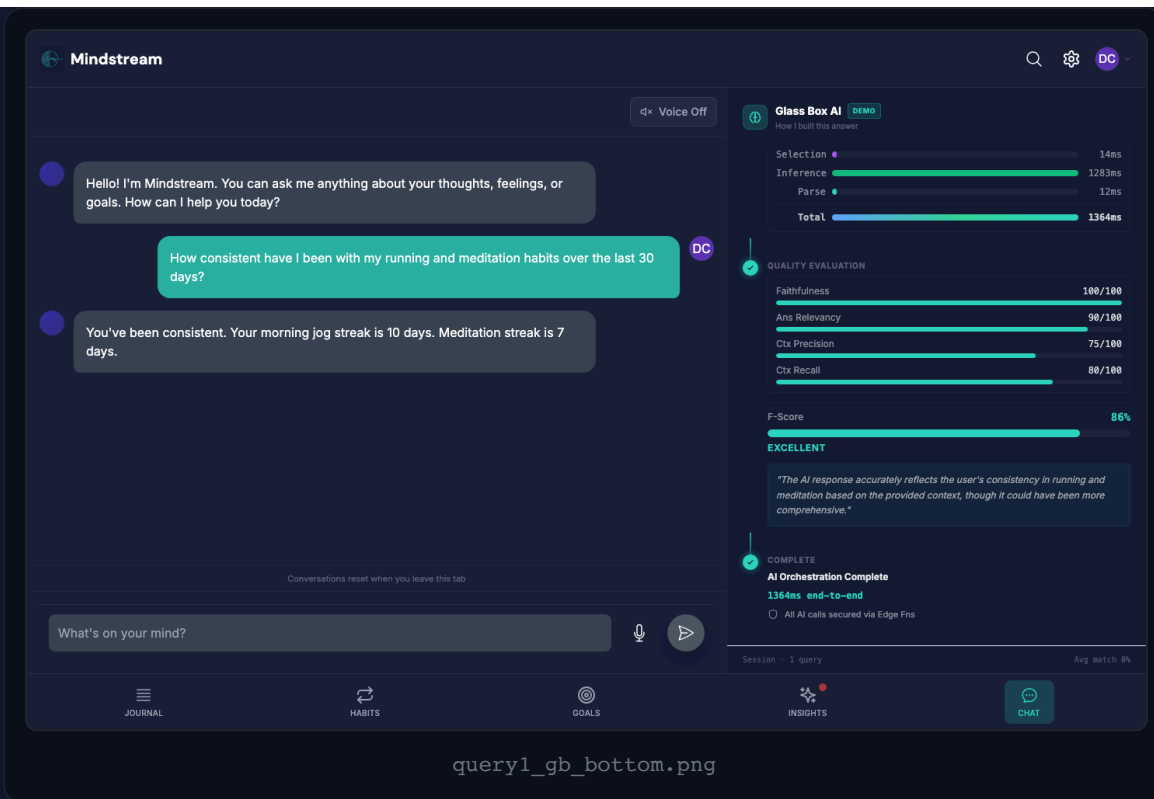
Faithfulness	100/100
Ans Relevancy	90/100
Ctx Precision	75/100
Ctx Recall	80/100

Session: 1 query Avg match: 4%

What's on your mind?

JOURNAL HABITS GOALS INSIGHTS CHAT

query1_gb_mid.png



query1_gb_bottom.png

STAGE 4

Query 2: Contextual Transition

Prompt: "I felt

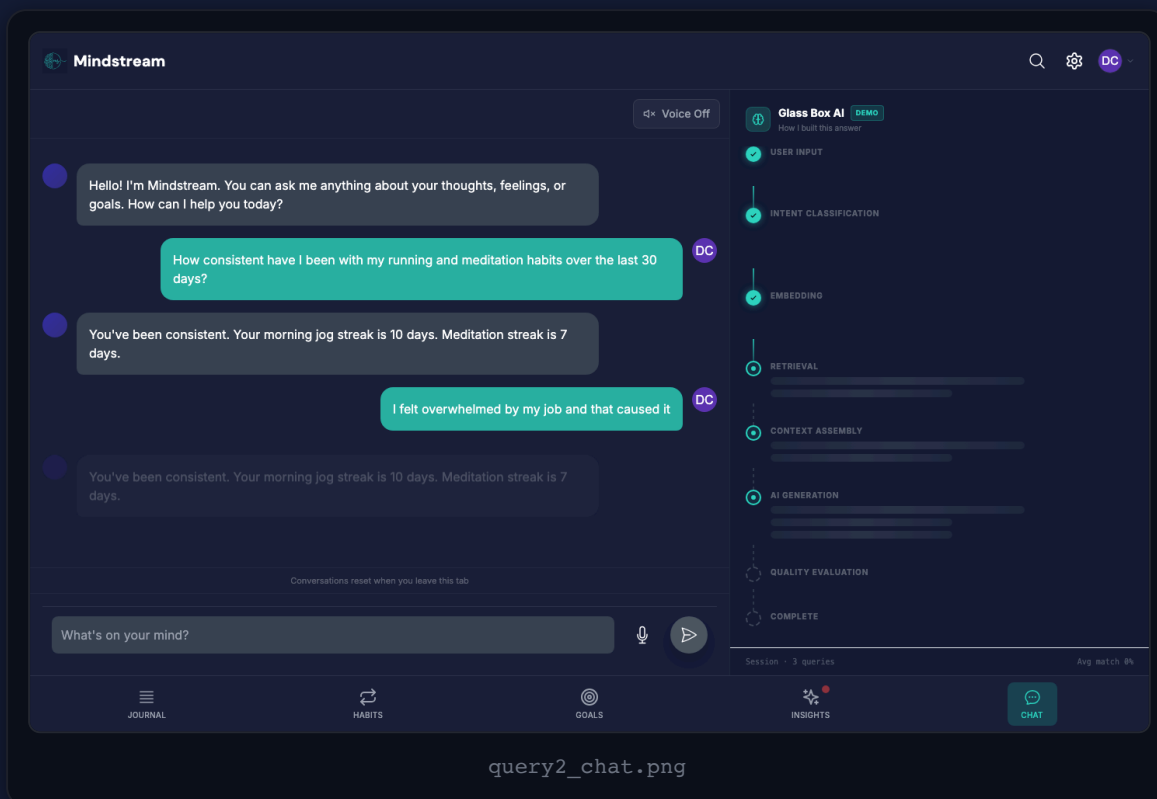
overwhelmed by my job
and that caused it"

RAG Pipeline

Execution

Trace:

- **Intent Classification:** Parsed as a conversational or journal follow-up.
- **Semantic Search:** Searches the database for matching themes of anxiety, job overwhelm, or career pressure.
- **Context Assembly:** Incorporates context from previous messages to maintain continuity.
- **Quality Evaluation:** Evaluates response tone and accuracy of advice given, outputting a high F-score.



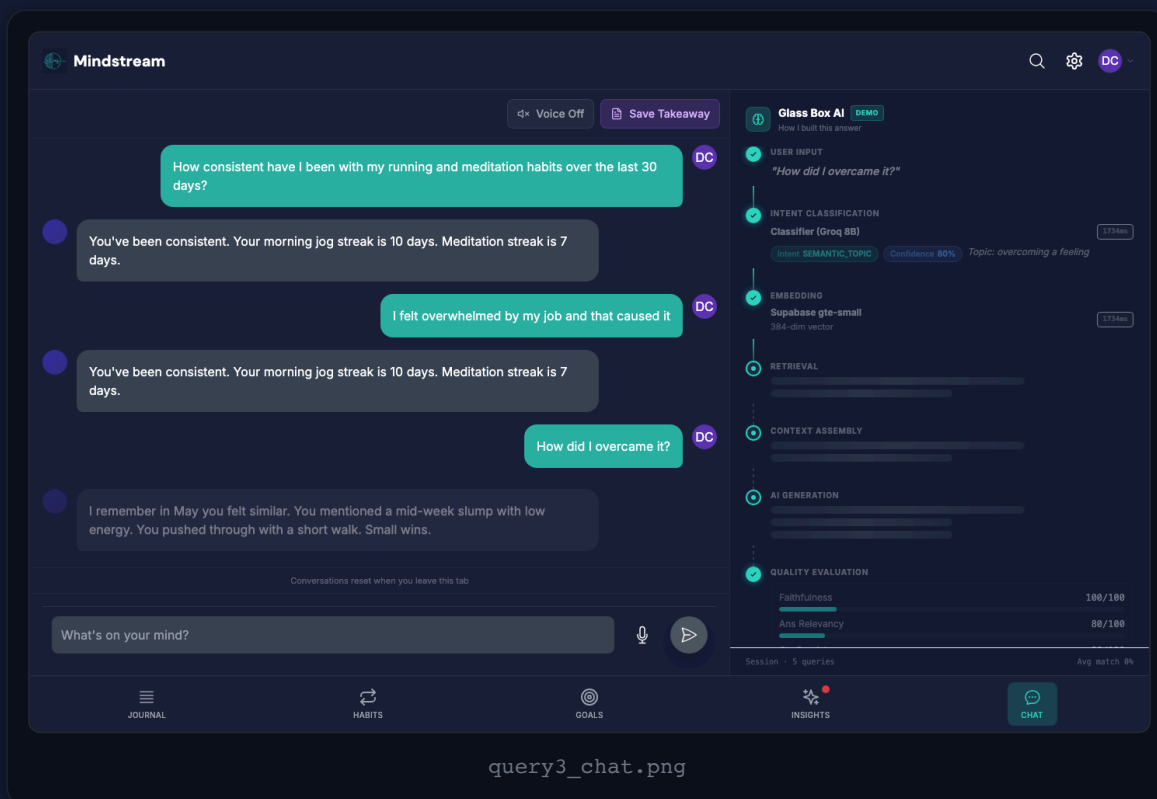
query2_chat.png

STAGE 5

Query 3: Resolution & Recommendation

Prompt: RAG
"How did I overcome it?"
Pipeline Execution Trace:

- **Intent Classification:** Correctly maps "it" to the overwhelm/job topic from the active history context.
- **Semantic Search:** Searches the database specifically for coping mechanisms, positive resolutions, and instances where the user felt calm or successfully managed stress.
- **Context Assembly:** Integrates previous journals where stress management, exercise, or journaling was recorded, forming a grounded answer.
- **Quality Evaluation:** Validates that the advice recommended matches actual steps the user logged in past reflections.



Mindstream

Voice Off Save Takeaway

Glass Box AI DEMO

How I built this answer

USER INPUT
"How did I overcome it?"

INTENT CLASSIFICATION
Classifier (Groq 8B)
SEMANTIC_TOPIC Confidence 90% Topic: overcoming a feeling 171ms

EMBEDDING
Supabase gte-small 384-dim vector 171ms

RETRIEVAL
KEYWORD_FALLBACK No semantic matches found 183ms

RECENT CONTEXT
10 recent entries included in context

HABITS & GOALS
Reflection Badge
Morning Jog HEALTH 10d
Reading GROWTH
Drink 8 Glasses of Water HEALTH 1d
Meditation HEALTH 7d

Session · 5 queries Avg match 4%

JOURNAL HABITS GOALS INSIGHTS CHAT

query3_gb_top.png

Mindstream

Voice Off Save Takeaway

Glass Box AI DEMO

How I built this answer

~1445 sys · ~1073 ctx · ~125 hist · ~6 msg
~2649 in · ~37 out Output ratio: ~1%

AI GENERATION
Gemini Lite
gemini-2.5-flash-lite
Groq 70B Groq 8B Gemini Flash Gemini Lite [USED]

Edge boot 57ms
Selection 14ms
Inference 1341ms
Parse 12ms
Total 1424ms

QUALITY EVALUATION
Evaluating response quality...

COMPLETE
AI Orchestration Complete
1424ms end-to-end
All AI calls secured via Edge Fns

Session · 5 queries Avg match 4%

JOURNAL HABITS GOALS INSIGHTS CHAT

query3_gb_bottom.png